InternSync – Software Project Management Plan (SPMP)

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CSC 305 – Semester 1 – 2026/2027

Project title: InternSync – Cooperative Training Coordination and Tracking System
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> This Markdown file is the maintained version of the SPMP. `InternSync_SPMP-1.pdf` in this folder is version 1.0 as submitted on Oct 3, 2026. Changes after 1.0 are listed in the revision history and in Change_Requests.md.

Team Information

Team Member	ID	Date
Salam Alsaour (Team Leader)	2250009168	Oct 3, 2026
Khalid Alajmi	2250004309	Oct 3, 2026
Ali Alghafly	2250002460	Oct 3, 2026
Ahmed Kanaan	2250009170	Oct 3, 2026
Abdullah Abalhareth	2250005266	Oct 3, 2026
Rakan Alzahrani	2250000631	Oct 3, 2026

Revision History

Version	Date	Description	Author
1.0	Oct 3, 2026	First version submitted.	Group 3
1.1	Oct 13, 2026	CR-01: zero-cost hosting in the resource plan and a new free-tier risk. Milestone status updated (M1 done, M2 in progress).	Salam Alsaour
1.2	Oct 20, 2026	CR-05: one platform for many universities and companies. Registration with verification and the Platform Admin added to Increment 1 (T4); CSV import and coordinator review of vacancies removed. M2 status updated; new risk for the larger Increment 1; the permission risk now covers organizations.	Salam Alsaour
1.3	Oct 24, 2026	Milestone status updated: Increment 1 built and tested (M3 demo on Nov 1).	Salam Alsaour
1.4	Oct 26, 2026	Milestone status updated: Increment 2 built and tested early (M4 demo on Nov 15). CR-08: the development plan (FR-MAT-06) moved from T6 to T5.	Salam Alsaour
1.5	Oct 27, 2026	Milestone status updated: Increment 3 built and tested early (M5 demo on Nov 25); FR-ORG-07 delivered in T6 as the risk plan allowed.	Salam Alsaour
1.6	Oct 28, 2026	T7 started early as Increment 4: the Could functions F16 – F19 and NFR-PRV-04, the offline mode of CR-10, and the release checks (browser tests, ZAP scan, load test, held-out AI sets). M6 status updated: UAT prepared, to be run by the owner.	Salam Alsaour
1.7	Oct 29, 2026	T8 done early as Increment 5: final report, requirements status, user guide, demo script, presentation, and the submission files. M6 and M7 status updated; UAT postponed by the owner's decision of Oct 29 (risk of volunteers not available has occurred).	Salam Alsaour

1. Project Organization

1.1 Process Model

- **Plan-driven development with incremental delivery.** All activities (requirements, design, implementation, testing, documentation) are planned in advance with dated phases and deliverables, and progress is measured against the plan.
- **Why:** the course fixes the deadlines and the documents to be submitted, so requirements, design, testing, and documentation follow dated phases. An agile process such as Scrum relies on flexible time and scope, which does not match a fixed schedule with graded deliverables.
- **Incremental delivery:** implementation is split into three planned increments ordered by priority (Must, then Should, then Could), so a working, tested system exists early. If the schedule slips, lower-priority functions are dropped instead of moving the deadlines. A strictly sequential Waterfall process was rejected because it leaves testing until the end.
- **Change control:** after the SRS is approved, requirement changes are recorded and accepted only if they fit the schedule.
- **Quality practices:** Git feature branches and pull requests with at least one reviewer; automated tests run on every pull request; documentation (SRS, design, test report) is written alongside the work.

1.2 Organizational Structure

- **Team Leader (Salam Alsaour)** coordinates all work, maintains the schedule, and reports to the course supervisor.
- **Five sub-teams:** Frontend/PWA (Ali, Khalid); Backend and Database (Salam, Khalid); AI services (Salam, Ahmed), with Ahmed also responsible for deployment; UI/UX design (Rakan); QA and Documentation (Abdullah). Members may support other sub-teams when needed.
- **Reporting lines:** sub-team members report to the Team Leader; the Team Leader reports to the course supervisor at each milestone.
- **Decisions** are made by team vote at the weekly meeting, and the Team Leader breaks ties. Technical decisions inside a sub-team are made by that sub-team and shared with the whole team.

2. Team Roles and Responsibilities

Team Member	Role	Responsibilities
Salam Alsaour	Team Leader; backend and AI services developer	Plans and tracks the schedule, runs the weekly meeting, reports to the course supervisor, integrates the components; develops the matching engine and similarity detection; in the backend: weekly reports, evaluations, scheduled notifications, PDF/Excel export.
Khalid Alajmi	Frontend and backend developer	Backend: database schema, authentication and role-based access control, approval workflow, API for profiles and vacancies. Frontend: screens for reports, evaluations, and the public certificate page.
Ali Alghafly	Frontend developer	React PWA: role dashboards, forms, Arabic/English interface with RTL support, notifications on the client.
Ahmed Kanaan	AI services developer; deployment lead	Skill extraction from CVs, match explanation and development plan (working with Salam on the AI services); deployment of the system (Docker and the hosting services).
Abdullah Abalhareth	QA and documentation lead	Test plan and test cases, user acceptance testing, SRS and design documents, keeps the task board and the risk register up to date.
Rakan Alzahrani	UI/UX designer	Wireframes and design system in Claude Design + Figma Make, usability of key screens.

3. Activities and Tasks

- Plan the project and collect requirements (SRS)
- Design the system architecture, database (ERD), and API
- Design the user interface (Claude Design + Figma Make wireframes, Arabic/English, PWA)
- Develop the backend (authentication, vacancies, reports, evaluations, approval workflow, notifications)
- Develop the frontend (role dashboards, forms, PWA features)
- Develop the AI services (matching, explanation and development plan, similarity detection)
- Integrate components and export grade sheets (PDF and Excel)
- Test the system (unit, integration, and user acceptance testing) and fix defects
- Prepare final documentation and presentation
- Manage the project (weekly meetings, tracking, change and risk control)

4. Schedule

All dates are in 2026. The Gantt chart is Figure 10 of the diagram pack.

ID	Activity	Start	End	Duration	Depends on	Owner
T1	Plan project and collect requirements (SRS)	Sep 25	Oct 3	9 days	–	Salam, Abdullah
T2	System design (architecture, ERD, API specification)	Oct 4	Oct 15	12 days	T1	Khalid, Ahmed, Salam
T3	UI/UX design (Claude Design + Figma Make wireframes)	Oct 4	Oct 15	12 days	T1	Rakan, Ali
T4	Increment 1: registration and verification (universities, colleges, companies, Platform Admin), authentication, profiles and CV upload, vacancies, weekly reports, bilingual interface foundation	Oct 16	Nov 1	17 days	T2, T3	All developers
T5	Increment 2: matching, attendance, evaluations, approval workflow, dashboards (all Must functions)	Nov 2	Nov 15	14 days	T4	All developers
T6	Increment 3: similarity detection, notifications, exports, PWA (Should functions); the explanation and skill gaps were delivered in T5 (CR-08)	Nov 16	Nov 25	10 days	T5	All developers
T7	Testing and bug fixing (unit, integration, user acceptance); Could functions if time allows. Built as Increment 4 from Oct 28 (all Could functions, CR-10, release checks)	Nov 26	Dec 10	15 days	T6	Abdullah, all
T8	Final documentation and presentation. Done as Increment 5 on Oct 29	Dec 11	Dec 15	5 days	T7	Abdullah, Salam
T9	Project management and monitoring (weekly meetings, tracking, change and risk control)	Sep 25	Dec 15	82 days	–	Salam

5. Milestones

Milestone	Deliverable	Target date	Status
M1: Requirements baseline	SRS approved by the team and requirements frozen	Oct 3	Done (SRS v1.0 on Oct 10)

Milestone	Deliverable	Target date	Status
M2: Design complete	Architecture, ERD, API specification, and Claude Design + Figma Make wireframes reviewed and approved	Oct 15	Done Oct 23 (8 days late after CR-05): design document 1.2, API specification 1.1, ERD in four parts, 32 wireframes, SRS 1.3 submitted; merged to main
M3: Increment 1 delivered	Demo of F1, F2, F3, F5, F10 foundation; automated tests pass	Nov 1	In progress: built and tested Oct 24 (test report); demo on Nov 1
M4: Increment 2 delivered (all Must functions)	Demo of F4, F6, F7, F8, F9; every Must function passes its tests	Nov 15	In progress: built and tested Oct 26, ahead of plan (test report); demo on Nov 15
M5: Increment 3 delivered	Demo of F11–F15; Should functions pass their tests	Nov 25	In progress: built and tested Oct 27, ahead of plan (test report); demo on Nov 25
M6: Testing complete	Test report; user acceptance testing with at least 8 participants, 80% or more of scenarios completed without help	Dec 10	In progress: Increment 4 built and its release checks done Oct 28 (test report); UAT prepared (plan); sessions postponed by the owner on Oct 29 and still open
M7: Final delivery	Final system, documentation, and presentation	Dec 15	In progress: documents, user guide, demo script, and presentation ready Oct 29 (final report); demonstration on Dec 15

6. Resource Planning

6.1 People

- Team Leader / project manager (1)
- Backend and database developers (2)
- AI services developers (2, one shared with the backend)
- Frontend developers (2, one shared with the backend)
- UI/UX designer (1)
- Deployment lead (1, also an AI services developer)
- QA and documentation lead (1)
- At least 8 volunteer users for user acceptance testing (students, faculty, and company representatives)

6.2 Software and Tools

- Python and FastAPI (backend and AI services)
- React with Vite (PWA frontend)
- PostgreSQL with the pgvector extension
- Open-source multilingual embedding model (e.g., multilingual-e5 via sentence-transformers)
- Docker and Docker Compose
- GitHub (version control, Projects board, GitHub Actions for automated tests, nightly backups, and keep-alive checks)
- Claude Design + Figma Make (UI/UX design)
- VS Code, pytest, and Postman (development and testing)

6.3 Hardware, Hosting, Data, and Effort

- **Development:** team laptops running the whole system with Docker Compose.
- **Hosting (zero cost, CR-01):** Vercel Hobby for the PWA; Hugging Face Spaces free CPU Basic (2 vCPU, 16 GB RAM) for the API container with the embedding model and scheduler; Supabase Free for PostgreSQL

with pgvector and private file storage; Gmail SMTP for email. No paid plan or credit card is needed. Details are in the design document (Section 3).

- **Data:** sample and synthetic data: two universities with two colleges each and several companies (so that permission tests cross organizations), CVs, vacancies, and reports, including a labelled test set of at least 20 student profiles for matching and 10 seeded copied reports for similarity detection.
- **Effort:** about 8 hours per member per week over 10 weeks (6 × 8 × 10 ≈ 480 person-hours).

7. Risk Identification and Management

Risk	Probability	Impact	Mitigation
Schedule slips because of team availability	High	High	MoSCoW priorities; weekly tracking; drop Could functions first, then Should functions.
Late requirement changes and scope creep	Medium	High	Freeze requirements after the SRS; change control; changes accepted only if they fit the schedule.
Increment 1 grows after CR-05 (registration, verification, Platform Admin) and starts 5 days late	High	High	One verification-request flow serves students, coordinators, and companies; CSV import and vacancy review are removed; universities, colleges, and skills are seeded by script so the Platform Admin screens can follow; if Increment 1 slips, FR-ORG-07 and FR-ORG-09 (Should) move to Increment 3.
Team members unfamiliar with FastAPI, React, or pgvector	Medium	Medium	Short learning tasks during the design phase; tasks assigned by skill; code reviews and pair work.
Matching or similarity results are unreliable or hard to explain	Medium	High	Rule-based fallback; show matched and missing skills; advisory use only; test on a labelled sample.
Extracting skills from CVs in different formats	Medium	Medium	Start with PDF and DOCX text; let the student review and edit extracted skills.
Arabic (RTL) text in PDF exports	Medium	Medium	Test export early in Increment 1 with bundled Arabic fonts.
Permission errors expose another user's or another organization's data	Low	High	Central role-based access control; every query scoped by college or company in one place; automated permission tests with two colleges and two companies (FR-AUT-05).
Inconsistent deadline, reminder, or approval states	Medium	Medium	Model approvals as an explicit state machine and test every transition.
Integration problems between frontend, backend, and AI services	Medium	Medium	Agree the API specification first; generate the TypeScript client from the OpenAPI schema; integrate every week.
Free hosting tiers sleep, pause, or hit their limits (Spaces sleeps after 48 h idle; Supabase pauses after 7 days of low activity, 500 MB database, 1 GB files; email sending quotas)	Medium	Medium	Scheduled keep-alive check and nightly backup through GitHub Actions; keep demo data small; monitor sizes on the dashboard; fall back to Docker Compose on a laptop for UAT and the demo.

8. Project Monitoring and Control

- **Weekly check:** the team meets every week to compare completed tasks with the schedule (T1–T9) and the task board on GitHub Projects.
- **Track unfinished tasks:** tasks not finished in the planned week are flagged, given a new owner or date, and their effect on the next milestone is assessed.
- **Milestone reviews:** at the end of each increment the team holds a demo, lists defects, and decides which remaining Should and Could functions stay in the plan.
- **Change control:** every change request is recorded with its effect on schedule and scope in Change_Requests.md; the Team Leader approves it after a team vote.
- **Risk review:** the risk table is reviewed every week and updated with new risks and their status.

- **Quality control:** pull-request reviews, automated tests, and a test report before final delivery.
- **Reporting:** the Team Leader reports progress to the course supervisor at each milestone.